



Systematic Evaluation of Signal-to-Image Transformation Pipelines for Vision Transformer-based Sleep Stage Classification

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Why automate sleep staging?



Sleep disorders (insomnia, apnoea, narcolepsy) are widespread



Gold standard: manual scoring of polysomnography (PSG)



Slow and subjective: hours per night; experts disagree



Deep learning: CNN/RNN → Vision Transformers (ViTs)

2–4 h

to score one full-night recording

82.6%

inter-scorer agreement, experienced experts

50–70 M

US adults living with a sleep disorder

Research gap and objectives

The gap

- Prior ViT work uses spectrograms or raw signals only
- GASF, GADF and MTF never compared for ViTs
- Best sampling rate before imaging unknown
- Pre-training value on PSG images unclear

Four experiments

- 1 Image transformation: GASF, GADF, MTF, spectrogram
- 2 Sampling rate: 512 / 256 / 128 / 64 Hz, FFT vs DWT
- 3 Pre-trained vs from-scratch ViT, balanced vs imbalanced
- 4 Seven architectures incl. ResNet-50 baseline

Dataset: CAP Sleep Database

80

recordings (6 healthy,
74 patients)

80,667

30-second epochs

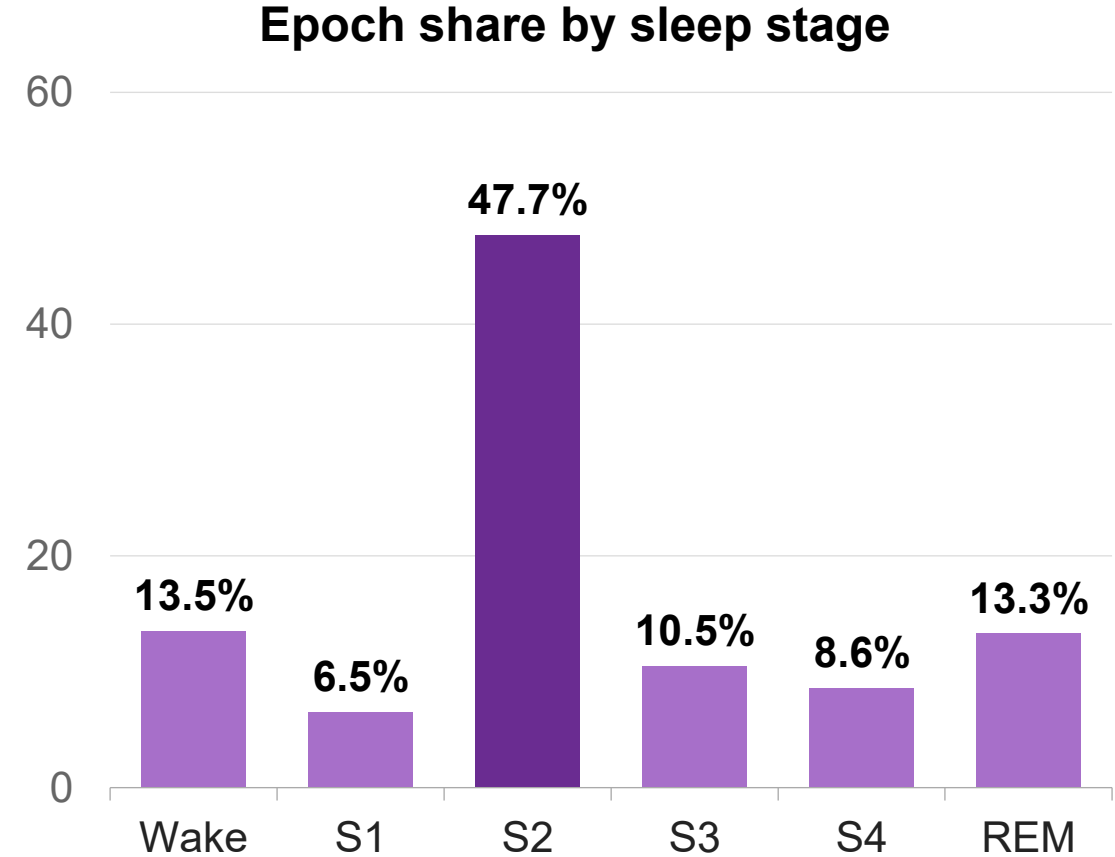
16

channels at 512 Hz

6

stages: Wake, S1–S4,
REM

Subject-wise 5-fold CV: no subject in both
train and test



Proposed Pipeline

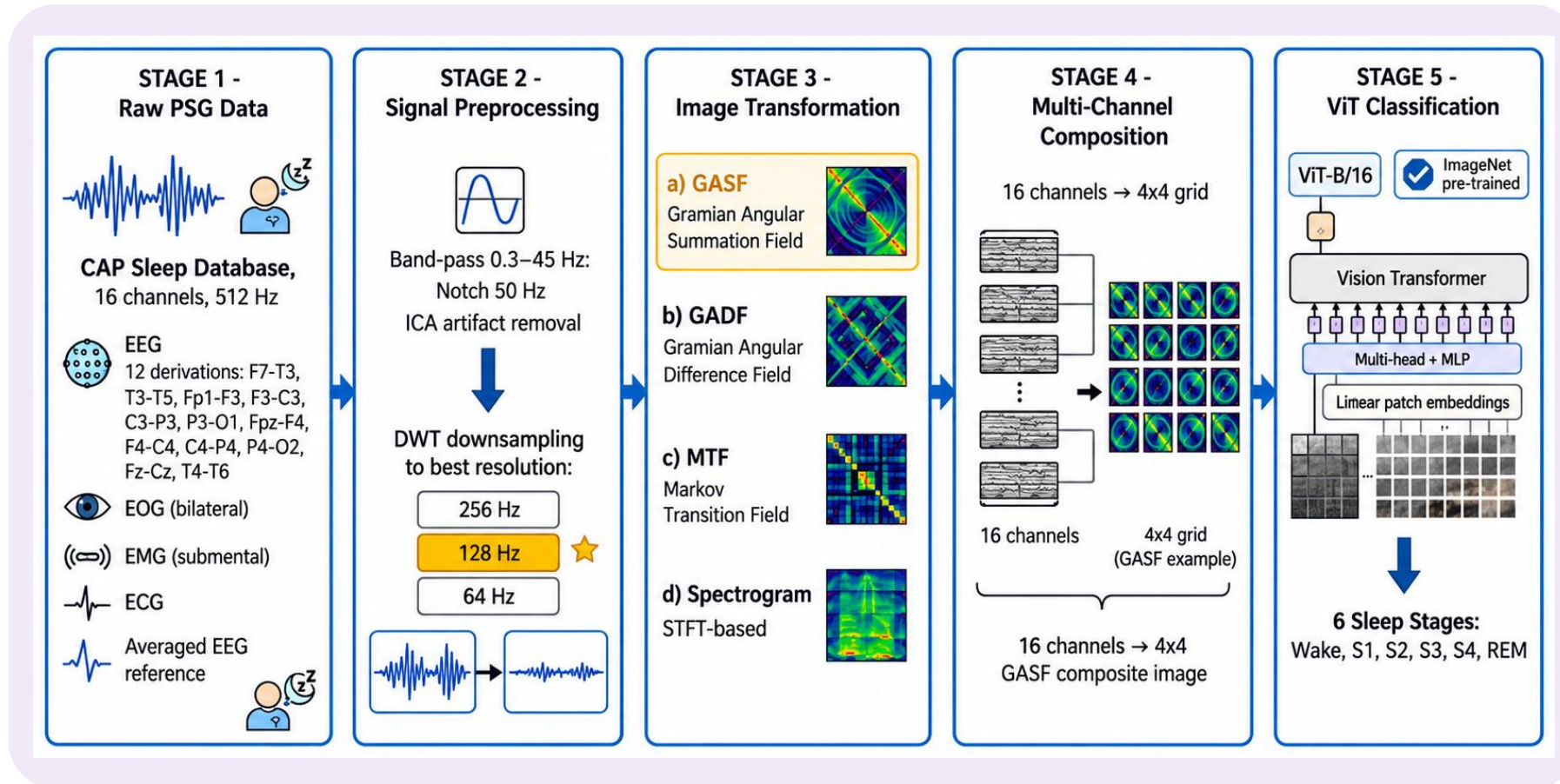


Figure 1. Overall five-stage pipeline

Four Signal-to-Image transformations



GASF

$$\cos(\varphi_i + \varphi_j)$$

Summed angles;
preserves global
temporal correlation



GADF

$$\sin(\varphi_i - \varphi_j)$$

Difference of angles;
less globally coherent



MTF

**Markov transition
probs**

32 bins; quantisation
loss, slow (5 s/epoch)



Spectrogram

STFT, Hamming 256

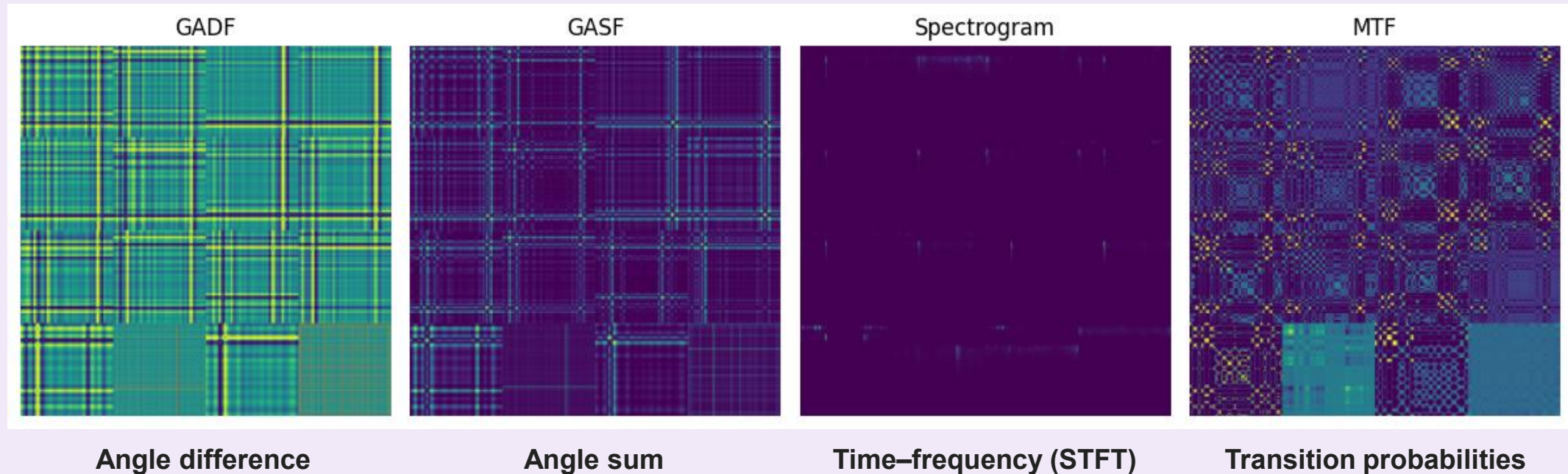
Fixed time–frequency
resolution



16 images → 4×4 grid → 224×224×3 → ViT-B/16

Same epoch, four images

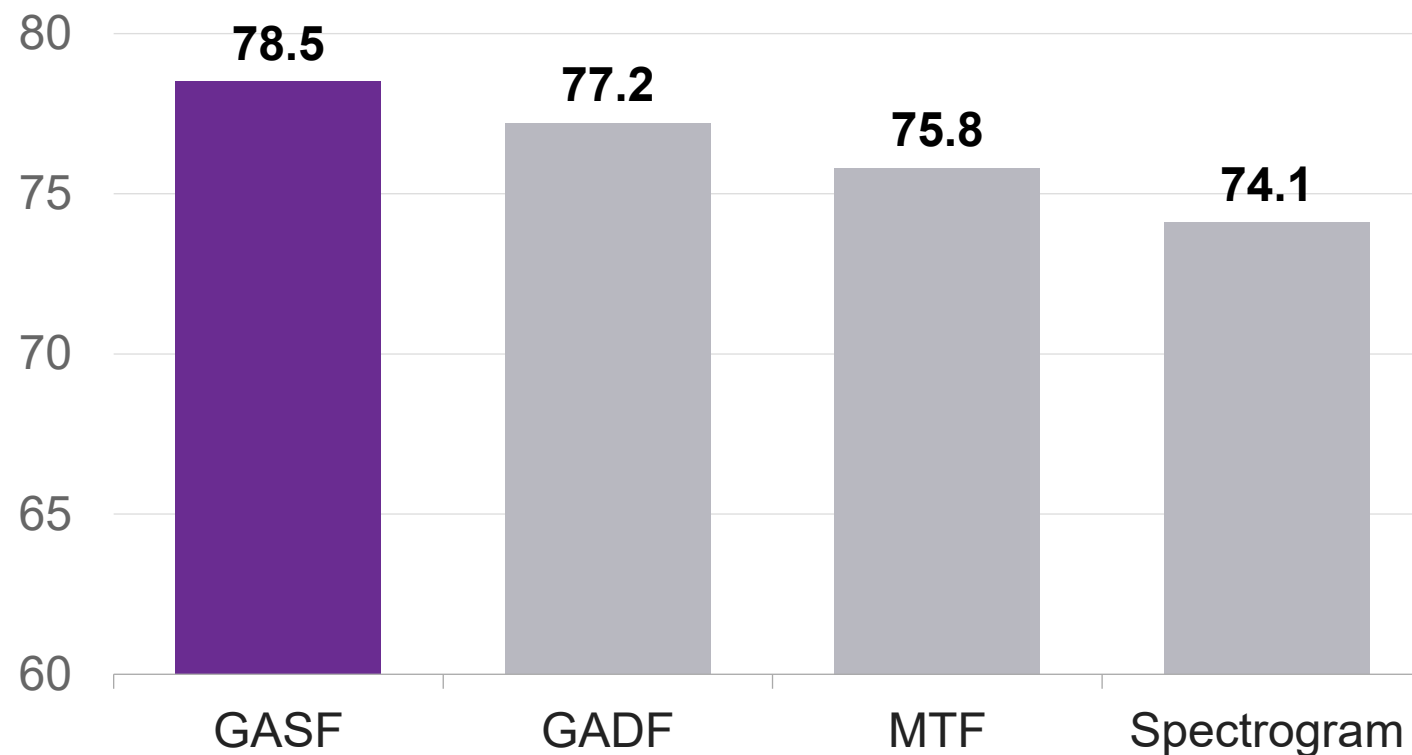
One subject, one 30 s epoch, original 512 Hz: 16 channels tiled into a 4×4 composite



⚡ The encoding changes the texture the ViT sees; GASF keeps rich structure

GASF performs best

EXPERIMENT ● ○ ○ ○



78.5%

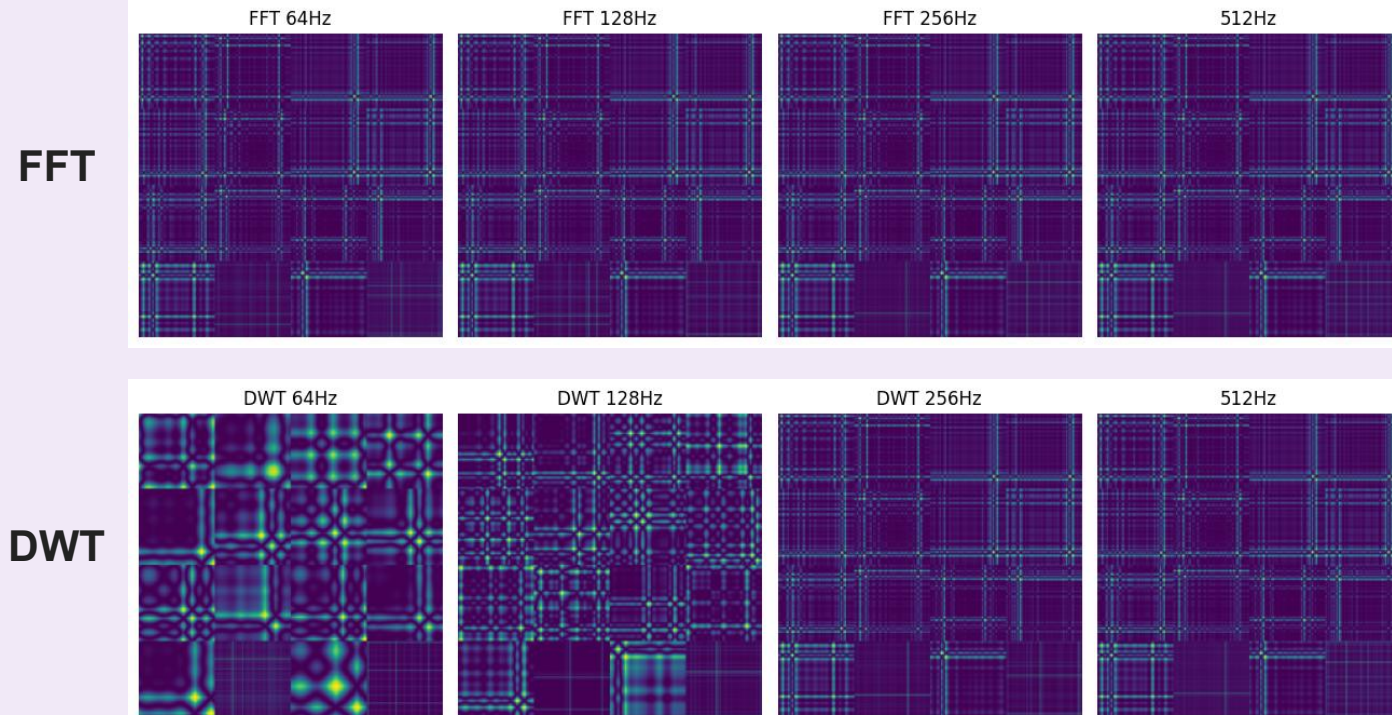
GASF accuracy (macro-F1
0.773)

5.0 s

MTF time per epoch vs 0.2 s for
GASF

⚡ GASF $\approx$ GADF; both clearly beat MTF and spectrogram

Downsampling: FFT vs DWT



What to look for

GASF images of the same epoch, resampled to 64, 128, 256 Hz and the 512 Hz original.

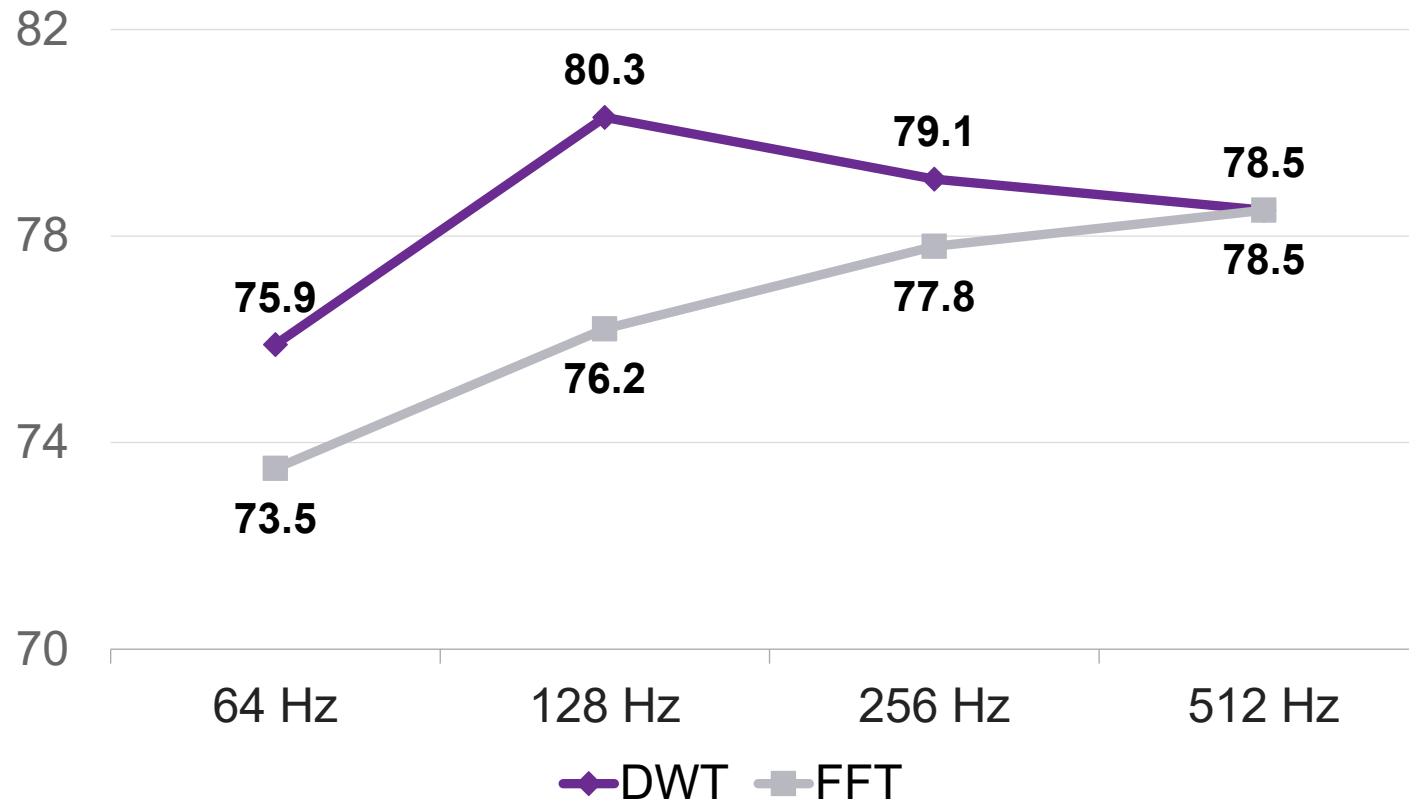
FFT: image structure stays almost the same at every rate.

DWT: images change visibly below 256 Hz; 128 Hz is the sweet spot for accuracy (next slide).

 Same epoch, same GASF encoding; only the resampling method differs

DWT at 128 Hz is optimal

EXPERIMENT ●●○○



80.3%

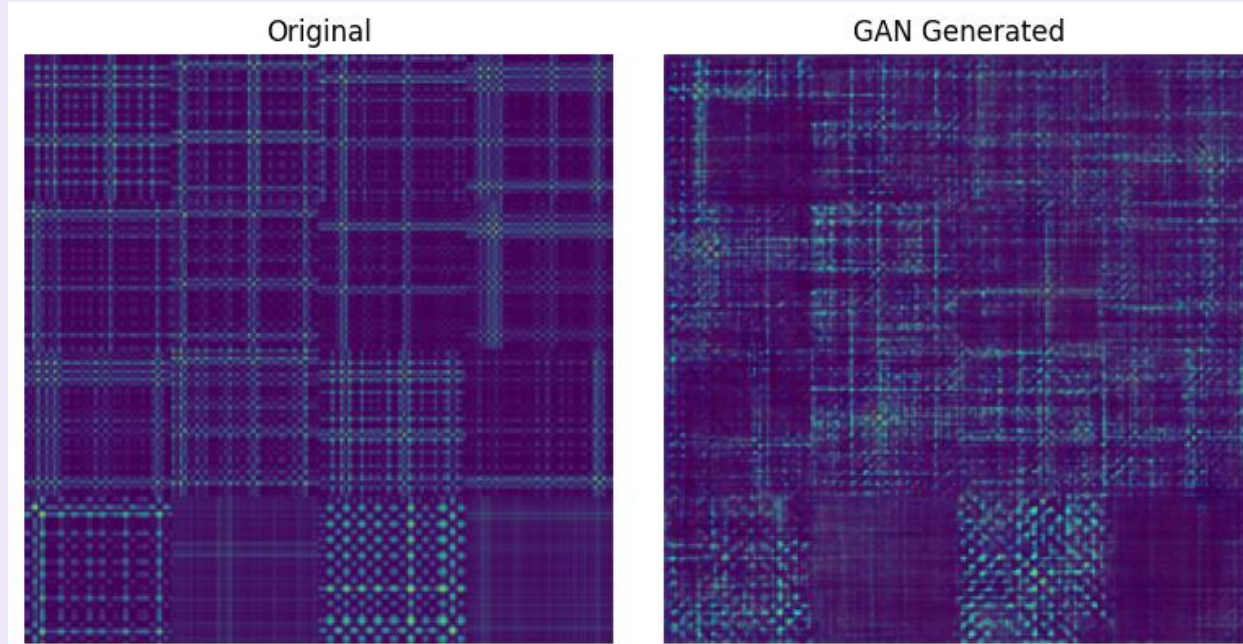
DWT @ 128 Hz (macro-F1
0.791)

+1.8 pp

over original 512 Hz signals

⚡ DWT beats FFT at every reduced rate; peak at 128 Hz

Class balancing with DCGAN



Real GASF composite vs DCGAN-generated sample, same class

How it works

A DCGAN learns the GASF composites of each under-represented class.

Synthetic images are added to training folds only.

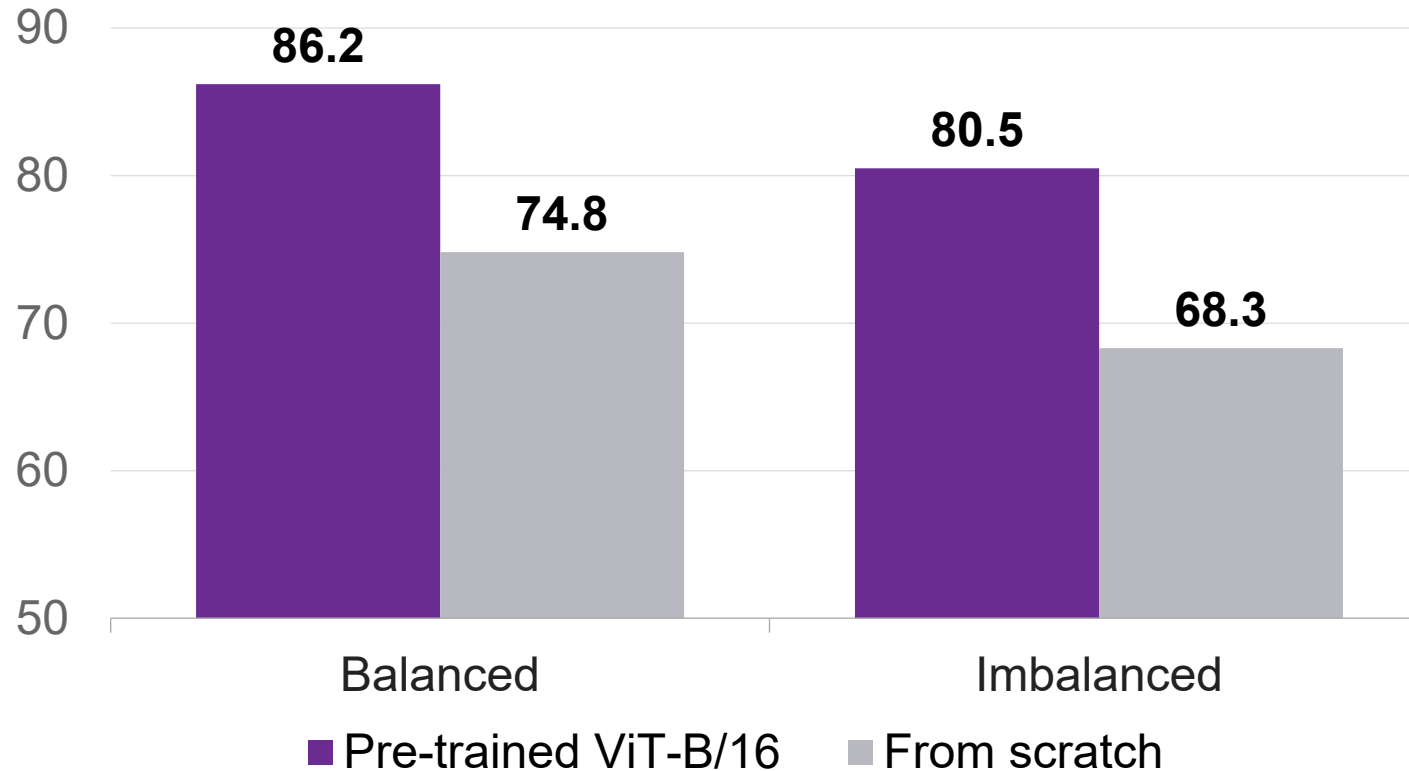
Test folds stay 100% real, so there is no leakage.



Synthetic samples keep the 4×4 channel layout and balance the classes

Pre-training adds 11+ points

EXPERIMENT ●●●○



+11.4 pp

balanced: 86.2% vs 74.8%

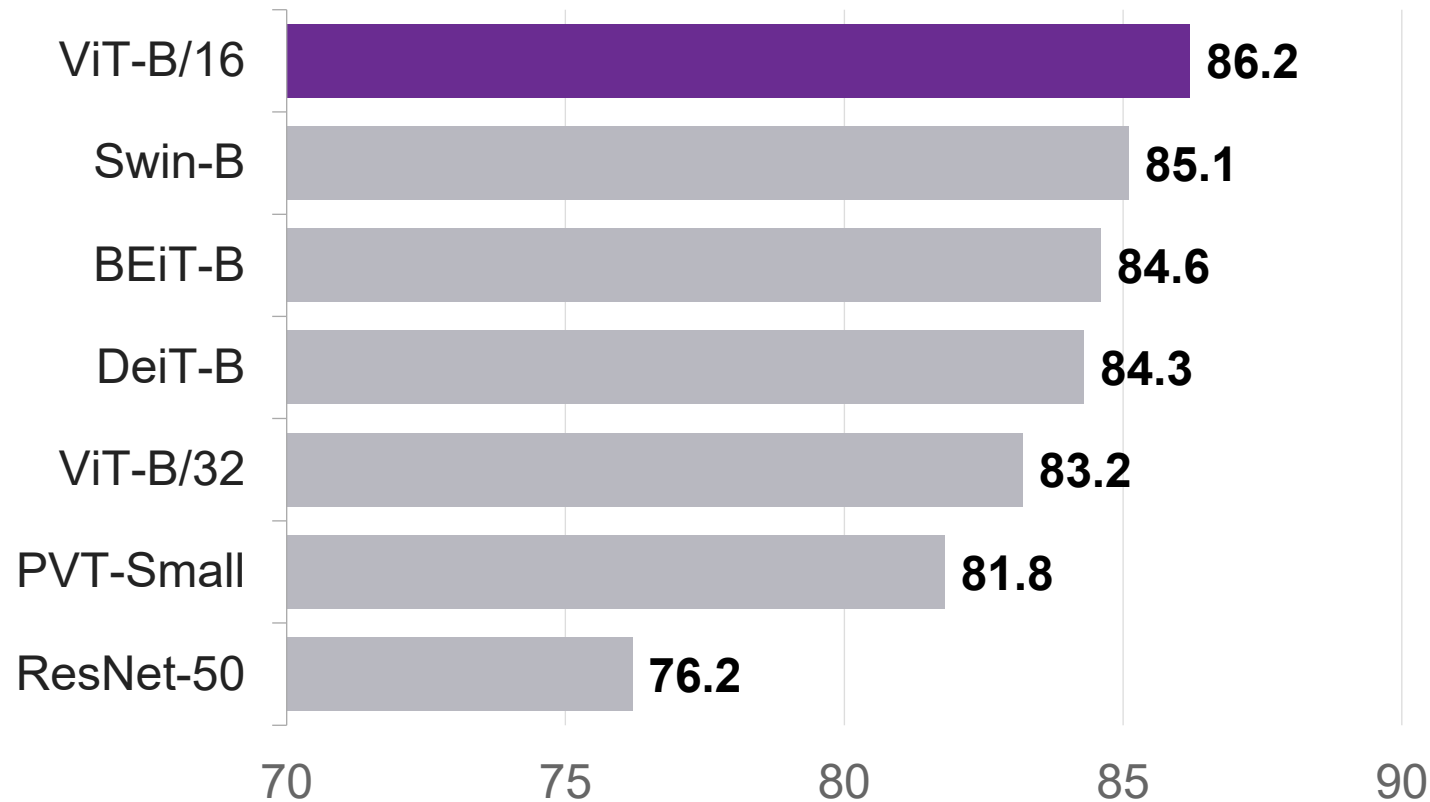
+12.2 pp

imbalanced: 80.5% vs 68.3%

⚡ Balancing (DCGAN, training folds only) adds ≈ 5.7 pp

ViT-B/16 beats other models

EXPERIMENT ●●●●



86.2%

ViT-B/16 (macro-F1 0.853)

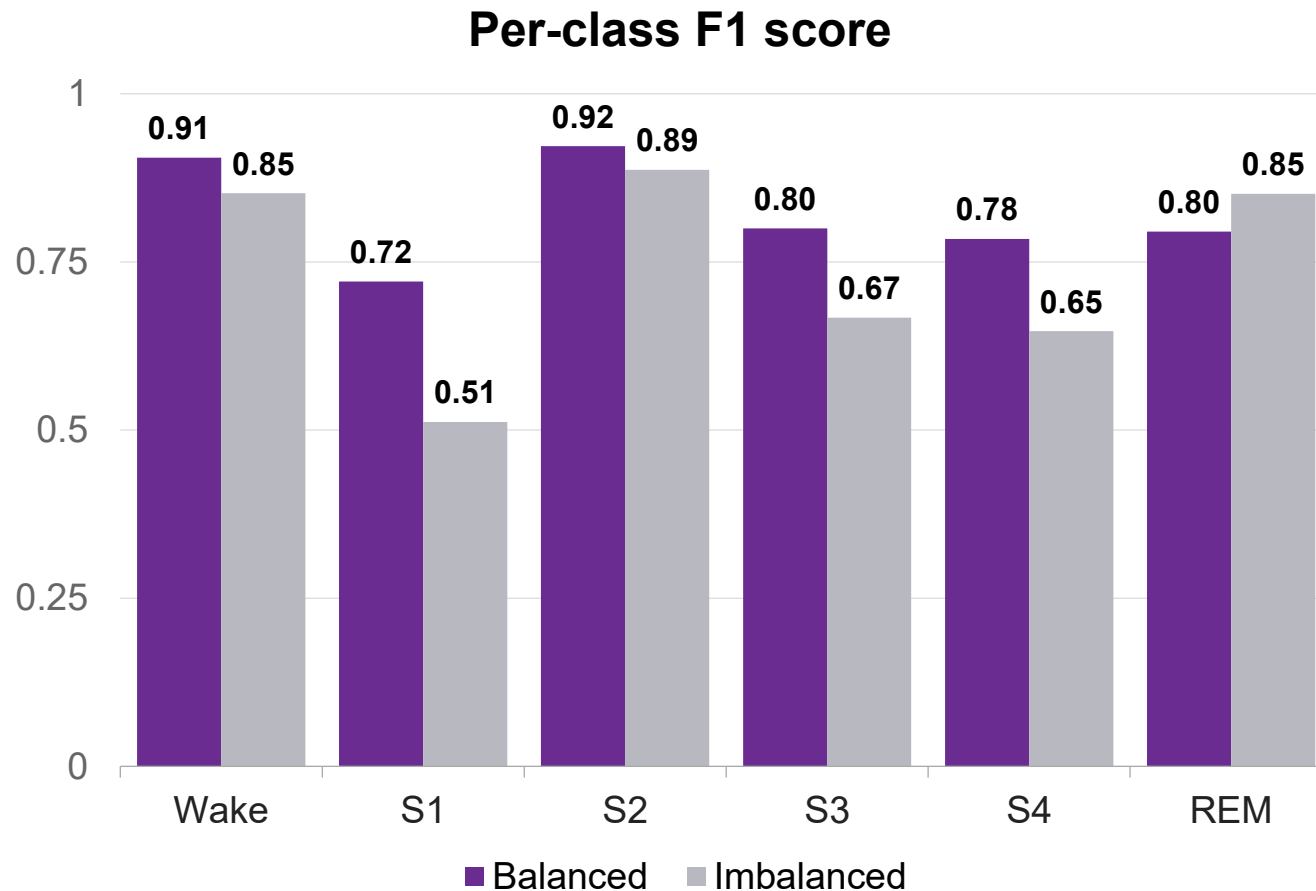
+10.0 pp

over ResNet-50 CNN baseline



Transformers outperform the CNN; PVT-Small is the efficient option

Class-wise results and limitations



Limitations

- S1 (6.5% of data) is weakest
- Single database; no cross-dataset test
- R&K labels, not AASM
- No formal significance tests
- 4×4 channel layout not ablated

Conclusions and future work

Best pipeline

-  **GASF** image encoding
-  **DWT** resampling to 128 Hz
-  **Fine-tuned** ImageNet ViT-B/16
-  **86.2%** accuracy, 0.853 macro-F1

Future work

-  Multimodal fusion
-  Lightweight ViTs for edge use
-  Cross-dataset validation

Thank you

Questions?

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